Band-1 Dirichlet / Piltz Divisor Problem — Next Exponent Probe (Collapse 1.13)

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Band-1 Dirichlet / Piltz Divisor Problem: Next Exponent Probe

Honesty stamp. This note **cites** the best published/arXiv **proven** upper bound on the error exponent α in

$$\Delta(x) = \sum_{n \le x} d(n) - x(\log x + 2\gamma - 1) = O_\varepsilon(x^{\alpha+\varepsilon})$$

and reports a **finite-range numerical illustration** of $\Delta(x)$ for selected $x \le 5 \cdot 10^7$. It does **not** claim a new proven exponent. Finite empirics are **not** proofs. **Not a full close. No Clay / ToE stamp. GATE-04 locked.**

Dirichlet / Piltz divisor exponent remains OPEN (conjecture $\alpha = 1/4$).

Abstract

Literature best proven α (2024–2026 check): Li–Yang (2023), $\alpha^* \approx 0.3144831759741$, improving Huxley (2003) $\alpha \le 131/416 \approx 0.314903846$. Bourgain–Watt's claimed $517/1648 \approx 0.31371$ (arXiv:1709.04340) was **withdrawn** (2023-07) and is **not** elevated.

Box: exact $\sum_{n \le x} d(n)$ via Dirichlet floor-blocks; $\Delta(x)$ for selected x; labeled **numerical illustration only**.

Problem (Band-1 collapse item H-1.13)

1. Cite the current **published** best α for $\Delta(x) = O_\varepsilon(x^{\alpha+\varepsilon})$ without invention.
2. Compute $\Delta(x)$ on a certified small–medium box window as a **numerical illustration**.
3. Keep GATE-04, Clay rhetoric, and full-close claims out of the claim surface.
4. Piltz k-divisor generalization noted only as sibling problem; tonight's target is $k = 2$ (Dirichlet).

Literature ledger (2024–2026 web/arXiv check; not re-proved on box)

Normalization: $\Delta(x) = O_\varepsilon(x^{\alpha+\varepsilon})$.

Year	α bound	Authors	Status
1849	$1/2$	Dirichlet	historical
1904	$1/3$	Voronoï	historical
1988	$7/22 \approx 0.31818$	Iwaniec–Mozzochi	published
1993	$23/73 \approx 0.31507$	Huxley	published
2003	$131/416 \approx 0.314903846$	**Huxley**	previous record
2017	$517/1648 \approx 0.31371$ (claimed)	Bourgain–Watt	**WITHDRAWN 2023** — not elevated

Year	α bound	Authors	Status
2023	$\alpha^* \approx 0.3144831759741$	**Li–Yang**	**best known**
—	1/4	Hardy conjecture	**OPEN** (optimal)

Primary citation (best known). Xiaochun Li, Xuerui Yang, *An improvement on Gauss's Circle Problem and Dirichlet's Divisor Problem*, arXiv:2308.14859 (2023). They prove $\Delta(X) = O_\varepsilon(X^{\alpha^*+\varepsilon})$ with $\alpha^* \approx 0.3144831759741$ the unique root in $[0.3, 0.35]$ of their defining equation (1.4) (same paper that improves the Gauss circle radius-form exponent to $2\alpha^*$).

Previous record. M. N. Huxley, *Exponential sums and lattice points III*, Proc. London Math. Soc. **(3) 87** (2003), 591–609: $\alpha \le 131/416$.

Bourgain–Watt hygiene. J. Bourgain, N. Watt, arXiv:1709.04340 claimed $\theta = 517/1648$ for the divisor/circle X-form; paper **withdrawn** by Nigel Watt (v2, Jul 2023) citing gaps in Propositions 2–3 / 1′. **Not** treated as the published record. (Related mean-square zeta work arXiv:1505.04161 is a different claim surface.)

Lower bound / conjecture. Hardy (1916): $\Delta(x) = \Omega(x^{1/4})$ (up to logs), so $\alpha \ge 1/4$; conjecture $\alpha = 1/4$. Soundararajan (2003) and later omega refinements (Sourmelidis; Mahatab 2026; sharp omega work 2026) improve Ω-side log factors but **do not** change the proven upper α.

Method barrier (context only). Bombieri–Iwaniec / Huxley-type methods face a widely noted obstacle near $5/16 = 0.3125$; Li–Yang sits just above that barrier. Closing to 1/4 needs new ideas.

2024–2026 lag note. Several 2024–2026 papers still quote Huxley 131/416 alone; the honest ledger elevates Li–Yang (2023) as the standing improvement (consistent with tonight's Gauss-circle Band-1 note).

Method (box)

1. **Exact sum.** $\displaystyle\sum_{n\le x} d(n) = \sum_{k\le x} \lfloor x/k \rfloor$ via Dirichlet floor-blocks in $O(\sqrt{x})$.
2. **Error.** $\Delta(x) = \sum d(n) - x(\log x + 2\gamma - 1)$ with γ Euler–Mascheroni (IEEE double for main term; sum exact as Python `int`).
3. **Selected x.** $10, 10^2, \ldots, 10^7, 5\cdot 10^7$ — sparse illustration, not a dense census.
4. **Unit tests.** $\sum_{n\le 10} d(n) = 27$; $\sum_{n\le 100} d(n) = 482$ (naive cross-check).
5. **Label.** Numerical illustration — **not** a proven exponent; descriptive θ_{fit} on sparse points is **not** comparable to α^*.

Results

Literature (cited, not re-proved)

Quantity	Value
Best published α	$\alpha^* \approx 0.3144831759741$ (Li–Yang 2023)
Previous	Huxley $131/416 \approx 0.314903846$
Bourgain–Watt 517/1648	**withdrawn** — not elevated
Conjecture	$\alpha = 1/4$ **OPEN**

Box $\Delta(x)$ (numerical illustration)

x	$\sum d(n)$	$\Delta(x)$	$\lVert\Delta\rVert/x^{1/4}$
10	27	+2.4298	1.3664
10^2	482	+6.0398	1.9100

x	$\sum d(n)$	$\Delta(x)$	$\|\Delta\|/x^{1/4}$
10^3	7069	+6.8134	1.2116
10^4	93668	+20.2830	2.0283
10^5	1166750	+14.3205	0.8053
10^6	13970034	+92.1122	2.9128
10^7	162725364	+94.1924	1.6750
$5 \cdot 10^7$	894098529	+284.3402	3.3814

Unit tests: **PASS**. Wall ≈ 0.002 s on this box.

Descriptive sparse log–log fit on $x \geq 10^3$: $\theta_{\text{fit}} \approx 0.324$ — **not a proof**; noise on eight points; must not be read as beating Li–Yang.

What this does / does not claim

Claim	Status
Cite best published α	**Done** (Li–Yang α^*)
Honest Bourgain–Watt withdrawal note	**Done**
Exact finite $\Delta(x)$ illustration	**Done**
New proven exponent	**No**
Full close / Clay / ToE	**No**
GATE-04	**LOCKED**

Paths

- Lab: `/workspace/mission-control/experiments/band1/dirichlet_divisor/`
- Results: `.../results.json`
- Paper: `/workspace/mission-control/papers/2026-09-11_Band1_Dirichlet_divisor_exponent_probe.md`
- PDF: `/workspace/DOWNLOAD_PDFs/2026-09-11_Band1_Dirichlet_divisor_exponent_probe.pdf`

References (key)

1. M. N. Huxley, *Exponential sums and lattice points III*, Proc. London Math. Soc. **(3) 87** (2003), 591–609.
2. X. Li, X. Yang, *An improvement on Gauss's Circle Problem and Dirichlet's Divisor Problem*, arXiv:2308.14859 (2023).
3. J. Bourgain, N. Watt, arXiv:1709.04340 — **withdrawn** Jul 2023.
4. G. H. Hardy, *On Dirichlet's Divisor Problem*, Proc. Lond. Math. Soc. (1916/17).
5. K. Soundararajan, *Omega results for the divisor and circle problems*, IMRN (2003).